



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES

Digital

Part A: Content Design

Course Title	Computer Vision
Course No(s)	AIML* ZG525 Computer Vision
Credit Units	4
Content Authors	Ms. Seetha Parameswaran
Version	1.0
Date	June 26 th 2023

Course Objectives

No	Course Objective
C01	Students should understand the fundamentals of a camera producing an image, including camera calibration, optical distortions, perspective corrections etc.
C02	Students should be familiar with various building block algorithms in Computer Vision, including Image processing and Deep Learning with emphasis on the algorithm building blocks.
C03	Students should create at least one end-user application.

Text Book(s)

T1	Szeliski, R., 2022. Computer vision: algorithms and applications. Springer Nature.
T2	Image Processing, Analysis, and Machine Vision: Milan Sonka, Vaclav Hlavac, Roger Boyle, Fourth edition, Cengage Learning

Reference Book(s) & other resources

R1	Forsyth, D. A., & Ponce, J. (2002). Computer vision: a modern approach. Second Edition. Prentice hall
R2	Practical Machine Learning for Computer Vision: End-to-End Machine Learning for Images, O'Reilly, 2021

Content Structure

- 1 Computer Vision (4 hrs)
 - 1.1 What is Computer Vision? (T1 Ch 1.1)
 - 1.2 Why Computer Vision is hard? (T2 Ch 1.2)
 - 1.3 Applications of Computer Vision (T1 Ch 1.1)
 - 1.4 Image representation and image analysis tasks (T2 Ch 1.3)
 - 1.5 Image digitization - Sampling and resolution (T2 Ch 2.2)
 - 1.6 Digital Images (T2 Ch 2.3)
 - 1.7 Digital Image types -Binary, Gray-scale and Color (Class Notes)
 - 1.8 Color Images (T2 Ch 2.4)
 - 1.9 Color spaces: RGB and HSV (T2 Ch 2.4)
- 2 Low-level Vision (3 hrs)
 - 2.1 Histogram and Histogram equalization (T1 Ch 3.1.4)
 - 2.2 Gray-scale transformation (T2 Ch 5.1.2)
 - 2.3 Image Smoothing (T2 Ch 5.3.1)
 - 2.4 Connected components in images (T1 Ch 3.3.4)
 - 2.5 Use case: Sharpening, blur, and noise removal using Filtering (T1 Ch 3.4.4)
- 3 Mid-level Vision (4 hrs)
 - 3.1 Edge Detection using Gradients, Sobel, Canny (T1 Ch4.2, T2 Ch 5.3.2, 5.3.5)
 - 3.2 Line detection using Hough transforms (T1 Ch 4.3, T2 Ch 5.3.10)
 - 3.3 Semantic information using RANSAC (T1 Ch 4.3, T2 Ch 10.3)
 - 3.4 Image region descriptor using SIFT (T2 Ch 10.2)
 - 3.5 Use case: Pedestrian detection Using HoG and SIFT descriptors and SVM (T1 - Ch 14.2)
- 4 Object Segmentation (4 hrs)
 - 4.1 Types of Segmentation: Semantic vs Instance (Class Notes)
 - 4.2 Segmentation using Agglomerative clustering, Kmeans (R1 Ch 9.3)
 - 4.3 Mean-shift clustering (T2 Ch 7.1)
 - 4.4 Vision Transformer (Class Notes)
 - 4.5 Popular DNN Architectures for Segmentation - Detectron family, SOLO, CondInst, Segment Anything Model (SAM), InternImage (Class Notes)
 - 4.6 Metrics for Object Segmentation (R1 Ch 9.5)
 - 4.6.1 mean IoU
 - 4.6.2 Pixel Accuracy,
 - 4.6.3 Boundary Error: ABPE, BDE
 - 4.7 Use cases for Object Segmentation - Crop classification from satellite imagery
- 5 Image Classification using Deep Learning (3 hrs)
 - 5.1 Pattern recognition methods in image understanding (T2 Ch 10.6), R1 Ch 15.3)
 - 5.2 Popular DNN Architectures: MobileNet, XceptionNet (Class Notes)
 - 5.3 Metrics for Image Classification (R1 Ch 15.1)
 - 5.3.1 Model Accuracy Metrics

- 5.3.1.1 Accuracy, Confusion Matrix, TPR, FPR, FNR, Top-K accuracy
 - 5.3.1.2 Precision, Recall, F1 Score
 - 5.3.1.3 AUC-ROC, AUC-PR
 - 5.3.1.4 Intersection-Over-Union (IoU)
 - 5.3.2 Model Performance Metrics
 - 5.3.2.1 FLOPs
 - 5.3.2.2 Memory Footprint for @ specific precision
 - 5.3.2.3 Inference Time on a specific hardware
 - 5.3.3 Metrics for Image Classification.
 - 5.3.3.1 Cross Entropy (Log Loss), Brier Score
 - 5.3.3.2 Macro-Precision, Macro-Recall, Macro-F1
 - 5.4 Example Use cases of Image Classification (R1 – Ch 15, 16)
 - 5.4.1 Automated sorting of fruits based on size, shape, color
 - 5.4.2 Apparel type classification from image
 - 5.5 Classifying Images Of Single Objects (R1 Ch16.2)
- 6 Object detection and Recognition (3 hrs)
- 6.1 Object detection (T2 Ch 9.2, R1 Ch 17.1)
 - 6.2 Mean-shift clustering (T2 Ch 9.2)
 - 6.3 Using YOLO (Class Notes)
 - 6.4 Metrics (class Notes)
 - 6.4.1 Average-Precision (AP)
 - 6.4.2 Mean-Average-Precision (mAP)
 - 6.5 Multi label object detection and recognition (Class Notes)
 - 6.5.1 Object Localization → Multilabel Classification
 - 6.5.2 Difference between Multiclass vs Multilabel Classification
 - 6.5.3 Popular Models: YOLO, SSD, Faster-RCNN
 - 6.6 Use case: Skin detection Ref: T1 – 14.1 and R1 - Ch17, 18
- 7 Object tracking (4 hrs)
- 7.1 Motion detection (R1 Ch 11)
 - 7.2 Tracking by Detection (R1 Ch 11.1)
 - 7.3 Tracking with the Mean Shift Algorithm (R1 Ch 11.2)
 - 7.4 Kalman Filters (R1 Ch 11.3)
 - 7.5 DNN architectures: DeepSORT, SiamFC, GSDT, SMILEtrack, SPARSEtrack
 - 7.6 Use case: Pedestrian tracking (add ref)
- 8 Visual Bag of Words and Semantic Hierarchy (4 hrs)
- 8.1 Knowledge representation (T2 Ch 9.1)
 - 8.2 Syntactic pattern recognition (T2 Ch 9.4)
 - 8.3 Scene labeling (T2 Ch 10.9)
 - 8.4 Semantic image segmentation and understanding (T2 Ch 10.10)
 - 8.5 Summarizing Images with Visual Words (R1 Ch 16.1.3)
 - 8.6 Application: Patch Classification in image of Breast Tumors Detection
- 9 Edge devices for computer vision (2 hrs)
- 9.1 ESP32 Cam module, Raspberry PI, Banana Pi etc
 - 9.2 Intel
 - 9.2.1 Core and Atom Processors

- 9.2.2 NUC
 - 9.2.3 Movidias VPUs
 - 9.2.4 OneAPI and OpenVino Libraries
- 9.3 Nvidia
 - 9.3.1 Jetson Platform - Nano, TX2, Orin
 - 9.3.2 DeepStream Library, CUDA, CUDNN
- 9.4 Others
 - 9.4.1 Google Coral

Optional Modules to be taken in Experiential Learning / Webinars / Tutorials / Assignments

- 1 Face detection and Recognition
 - 1.1 Boosting - Viola Jones algorithm (T2 Ch 10.7)
 - 1.2 DNN architecture: MTCNN, FastFace, RetinaFace
 - 1.3 Active appearance models (T2 Ch 10.5)
 - 1.4 Metrics (class Notes)
 - 1.4.1 IoU based metrics for Face Detection
 - 1.4.2 True Acceptance Rate (TAR), False AR, False Rejection Rate, TAR @ specific FAR, Top-K Identification Rate
 - 1.5 Use case: Attendance system on face image (add ref)
- 2 Optical Character Recognition
 - 2.1 Main challenges in OCR
 - 2.2 Popular Approaches for OCR:
 - 2.2.1 Edge and Contours based layout detection
 - 2.2.2 LayoutLM, DiT
 - 2.2.3 LSTM, YOLO
 - 2.2.4 Tesseract, EasyOCR
 - 2.3 Metrics for OCR
 - 2.3.1 Accuracy: character, word, sentence level
 - 2.3.2 String edit distance
 - 2.3.3 mAP for text localization
 - 2.4 Example Use cases of OCR
 - 2.4.1 Vehicle Number Plate recognition
 - 2.4.2 Invoice Parsing

Detailed Plan for Lab work

Lab No.	Lab Objective	Module Reference
1	Reading images Displaying images Color space conversion	1
2	Histogram equalization Gray-scale transformation Filtering applications like sharpening, blur, noise removal, smoothing	2
3	Edge detection using Sobel and Canny Line detection using Hough Transform RANSAC for semantic information SIFT image descriptor Pedestrian detection using HoG and SIFT	3
4	Image segmentation using Kmeans Mean-shift clustering for segmentation Vision transformer for segmentation Crop classification using satellite images	4
5	Fruit sorting using transfer learning Apparel type classification using transfer learning Comparison on metrics for evaluation (demo)	5
6	Mean shift clustering for object detection Object detection using Yolo and Faster RCNN Skin detection	6
7	Mean shift algorithm for object tracking Kalman filtering for object tracking Pedestrian tracking	7
8	Patch classification in images	8

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
EC-1(a)	Quizzes	Online		10%	
EC-1(b)	Assignments	Take Home		20%	
EC-2	Mid-Semester Test	Closed Book		30%	
EC-3	Comprehensive Exam	Open Book		40%	

Note:

Syllabus for Mid-Semester Test (Closed Book): Topics in Session Nos. 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)

Important links and information:

Elearn portal: <https://elearn.bits-pilani.ac.in> or Canvas

Students are expected to visit the Elearn portal on a regular basis and stay up to date with the latest announcements and deadlines.

Contact sessions: Students should attend the online lectures as per the schedule provided on the Elearn portal.

Evaluation Guidelines:

- 1 EC-1 consists of two Quizzes. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
- 2 EC-2 consists of either one or two Assignments. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
- 3 For Closed Book tests: No books or reference material of any kind will be permitted.
- 4 For Open Book exams: Use of books and any printed / written reference material (filed or bound) is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
- 5 If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam which will be made available on the Elearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course hand-out, attend the online lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the hand-out.